Requirements Analysis

Jens Bouma
Nicolas Llop Prieto
Husna Keskin
Ben Wierenga
Matz Leeflang

Version Control

Version	Date	Description
0.1	10-09-2024	Initial document
0.2	10-09-2024	Added requirements
0.3	13-09-2024	Added more requirements
0.4	03-12-2024	Languages fixes
0.5	18-12-2024	Changed layout document
0.6	07-01-2025	Making document ready

Table of Contents

1. Introduction 1

2. Requirements 2

2.1. Functional Requirements 2

2.2. Non-functional Requirements 5

2.3. Third-Party Requirements 6

2.3.1. Google Analytics 6

2.3.2. GitLab CI/CD pipeline 6

Bibliography 8

1. Introduction

The Requirements Analysis document serves as a detailed guide for the development of the Keuzewijzer outlining the functional, non-functional, user interface and third-party requirements that will guide the design, implementation, and testing phases of the project. The primary goal of this document is to define the expectations, constraints, and criteria for the system to ensure that all stakeholders are aligned and that the final product meets the specified needs.

For the traceability of the requirements throughout the process, the following explains how this will be achieved. Data carriers in the documentation will be provided with an identifier.

Identifier	Description
FR<#>	Functional Requirement
NFR<#>	Non-Functional Requirement
TRR<#>	Third-Party Requirements

2. Requirements

All the requirements are formatted in a table. The table consist of an identifier to track the requirements across documents. For example, the Functional Design and the Test report. A description of the requirement and prioritized using the MoSCoW-method (ProductPlan, 2025).

2.1. Functional Requirements

Identifier	Description	MoSCoW
FR0.1	The prototype is successfully deployed to a live or staging environment	Must
FR0.2	The prototype is accessible via a URL or other access points	Must
FR0.3	The prototype works on multiple devices	Should
FR0.4	Deployment should include documentation or instructions for testing	Should
FR1.1	The system shows the available graduate profiles	Must
FR1.2	The system ensures the list is updated with current profiles	Must
FR1.3	The system keeps showing the graduate profiles after selection	Must
FR2.1	When opening the webpage, a list of years should be shown for selection	Must
FR2.2	The current selected study year can be changed	Must
FR2.3	If a study year does not meet new curriculum, it should open the old Keuzewijzer	Must
FR3.1	The webpage should show a list of graduate profiles that can be selected	Must
FR3.2	The current selected graduate profile can be changed	Must
FR4.1	Show a path that shows the study year	Must
FR4.2	Each study year has 2 sections where a cluster can be followed	Must
FR4.3	If there are multiple clusters, they should be stacked and selectable	Should
FR4.4	Shows free choice if available	Should
FR4.5	Propaedeutic should be fixed	Must
FR5.1	If a link is available for the cluster, a link button should be shown	Must
FR5.2	A new window should open when clicking on the information	Must
FR6.1	The system should allow students to select a cluster during the free choice year	Must
FR6.2	Available clusters should be displayed in a list	Must

FR6.3	Students should be able to view prerequisites for each cluster	Should
FR6.4	The system should notify students of cluster conflicts with program requirements	Should
FR6.5	The selected choice should be able to be changed easily	Should
FR7.1	The system should allow students to retake a failed cluster	Must
FR7.2	If the retaken cluster can be followed the next semester, it should be selectable	Must
FR7.3	If retaking isn't possible in the next semester, it should become a free choice spot	Must
FR7.4	The system should notify students if retaking a cluster is not possible	Should
FR8.1	There should be an input for the username/email of the administrative account	Must
FR8.2	There should be an input for the password	Must
FR8.3	There should be a button to login	Must
FR8.4	Inputs should be validated for SQL or code injections	Must
FR8.5	A message should be shown for invalid credentials	Must
FR8.6	Login should only succeed if username and password match stored credentials	Must
FR8.7	There should be a "Forgot password?" link to recover login details	Should
FR8.8	The system should lock the account after 5 failed login attempts	Should
FR8.9	After successful login, the user should be redirected to the dashboard	Must
FR8.10	Password should be stored securely using hashing/encryption techniques	Must
FR8.11	The session should expire after a set period of inactivity (e.g., 30 minutes)	Should
FR8.12	Login process should work across all supported browsers and devices	Should
FR9.1	There should be a "Forgot Password?" link on the login page	Must
FR9.2	The link should direct the user to a password recovery form	Must
FR9.3	There should be an input field for entering the username	Must
FR9.4	An email should be sent with a secure token for password reset	Must
FR9.5	The token should expire after a predefined period	Must

FR9.6	If the user enters an invalid email, an error message should appear	Must
FR9.7	The system should validate the token before allowing a reset	Must
FR9.8	There should be input fields to enter and confirm the new password	Must
FR9.9	The new password should follow defined complexity rules	Should
FR9.10	A success message should be shown after a successful reset	Should
FR10.1	The system should display a list of graduate profiles	Must
FR10.2	The maintainer can add a new graduate profile via a form	Must
FR10.3	The maintainer can edit existing graduate profiles	Must
FR10.4	The maintainer can delete a profile with confirmation	Must
FR10.5	The system should validate data before saving changes	Should
FR10.6	All operations should be logged for auditing purposes	Should
FR11.1	The system should display a list of available clusters for each graduation profile.	Must
FR11.2	The maintainer can add a new cluster to a graduation profile.	Must
FR11.3	The maintainer can edit an existing cluster.	Must
FR11.4	The maintainer can delete a cluster with confirmation.	Must
FR11.5	The system should validate cluster data before saving changes.	Should
FR11.6	All operations should be logged for auditing purposes.	Should
FR12.1	The system should allow the maintainer to define prerequisites for each cluster.	Must
FR12.2	The system should allow the maintainer to associate clusters with specific semesters or study years.	Must
FR12.3	The system should allow the maintainer to define available clusters for free choice years.	Must
FR12.4	The system should allow the maintainer to configure the status of each cluster (active, inactive, etc.).	Must
FR12.5	The system should notify students if a selected cluster conflicts with any program requirements.	Should

FR12.6	The system should allow for easy navigation and filtering of clusters in the dashboard interface.	Should
FR13.1	The system should allow the maintainer to manage user roles (e.g., student, maintainer).	Must
FR13.2	The system should allow the maintainer to assign roles to different users.	Must
FR13.3	The system should restrict access based on user roles (e.g., students cannot modify profiles or clusters).	Must
FR13.4	The system should allow the maintainer to deactivate or delete user accounts.	Should
FR13.5	The system should log all user role changes and maintain an audit trail for security and accountability.	Should
FR14.1	The system shows the graduation profiles from Windesheim Almere	should
FR14.2	The system shows the clusters of Windesheim Almere	should
FR15.1	The system shows the graduation profiles of part-time Windesheim Zwolle	should
FR15.2	The system shows the cluster of part-time Windesheim Zwolle	should

2.2. Non-functional Requirements

Identifier	Description	MoSCoW
NFR1.1	A graduate profile should have its own color for visual separation	Must
NFR1.2	Each graduate profile should have its own unique color	Must
NFR1.3	The path matches the color of the selected graduate profile	Must
NFR1.4	When the last study year is finished, it shows a graduate hat icon	Should
NFR1.5	If there is no link available, it should show a grayed-out link button	Should
NFR1.6	The interface should display a loading indicator while the login request is processed	Could
NFR1.7	Load all components (e.g., graduation directions and course data) in under 3 seconds.	Must
NFR1.8	Support up to 500 simultaneous users without degradation.	Must
NFR1.9	Authentication via the school's student portal.	Could
NFR1.10	Ensure compliance with GDPR for data protection.	Must

NFR1.11	Encrypt data transmissions using SSL/TLS.	Must
NFR1.12	Intuitive and user-friendly design with minimal learning curve.	Must
NFR1.13	Mobile responsiveness and adherence to WCAG 2.1 for accessibility.	Must
NFR1.14	System must accommodate future courses and graduation paths without redesign.	Must
NFR1.15	Maintain 99.9% availability during the academic year.	Must
NFR1.16	Ensure regular backups of user data.	Must

2.3. Third-Party Requirements

2.3.1. Google Analytics

Identifier	Description	MoSCoW
TRR1.1	Integrate Google Analytics into the project to track user interactions, page views and other metrics.	Should
TRR1.2	Ensure Google Analytics is implemented with a GDPR popup.	Should

2.3.2. GitLab CI/CD pipeline

Identifier	Description	MoSCoW
TRR2.1	The CI/CD pipeline is set up using appropriate tools (e.g., Jenkins, GitLab CI, CircleCI) that align with our project needs.	Must
TRR2.2	The pipeline includes automated build processes that trigger on code commits to the repository.	Must
TRR2.3	Automated testing (unit tests, integration tests) is integrated into the pipeline to ensure code quality before deployment.	Must
TRR2.4	The deployment process to staging and production environments is automated, reducing manual steps and potential errors.	Must
TRR2.5	Notifications are configured to alert the team of build and deployment statuses, including failures and successes.	Must

TRR2.6	Documentation for the CI/CD pipeline is created, including setup instructions, workflows, and troubleshooting guidelines.	Should
TRR2.7	The pipeline is regularly reviewed and optimized based on team feedback and project needs.	Should

Bibliography

ProductPlan. (2025). *MoSCoW Prioritization*. Opgehaald van ProductPlan: https://www.productplan.com/glossary/moscow-prioritization/